Distorted-Wave Born Approximation (DWBA) for Single-Nucleon Transfer Reactions

Method documentation for `nexus-nrs`

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Contents

1 Overview **3**

2 Kinematics **3**
2.1 Laboratory–to–CM energy . . . 3
2.2 Reduced mass and mass factor . . . 3
2.3 Wave-numbers . . . 3
2.4 Q-value and exit-channel energy . . . 4

3 Bound-State Wave Function **4**
3.1 Radial Schrödinger equation . . . 4
3.2 Numerov integration . . . 4
3.3 Binding-potential . . . 4
3.4 Normalization . . . 5

4 Optical-Model Potential **5**
4.1 Volume terms . . . 5
4.2 Surface imaginary term . . . 5
4.3 Spin-orbit term . . . 5
4.4 Coulomb potential . . . 6
4.5 Parameters for $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$ . . . 6

5 Distorted Waves **6**
5.1 Radial equation and Numerov integration . . . 6
5.2 Normalization modes . . . 6

6 S-Matrix and Phase Shifts **7**
6.1 R-matrix . . . 7
6.2 Nuclear S-matrix . . . 7
6.3 Coulomb phase shifts . . . 7
6.4 Sommerfeld parameter . . . 8

7 Elastic Scattering **8**
7.1 Nuclear partial-wave bracket 8
7.2 Total elastic amplitude 8
7.3 Differential cross-section 8

8 DWBA Transfer Amplitude: Zero-Range Approximation **9**
8.1 Zero-range D0 approximation 9
8.2 Form factor 9
8.3 ZR radial integral 9
8.4 Angular-momentum selection rules 9
8.5 Reduced amplitude (Austern Eq. 5.6) 9
8.6 Transfer amplitude 10
8.7 Differential cross-section 10

9 Specific Reaction: $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$ **10**
9.1 Channels 10
9.2 Transferred nucleon 10
9.3 Partial-wave summation 10
9.4 Summary of the calculation chain 11

10 Numerical Parameters **11**

11 Code–Formula Correspondence **11**

1 Overview

This document describes the formulas implemented in the `dwba.transfer` namespace of `nexus-nrs`, with particular emphasis on the one-nucleon stripping reaction $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$. The principal ingredients are:

1. Non-relativistic kinematics (CM energies, wave-numbers, Q-value).
2. Bound-state wave function of the transferred nucleon.
3. Complex optical-model distorted waves for both channels.
4. S-matrix extraction, Coulomb phase shifts, and elastic scattering.
5. Zero-range DWBA (ZR-DWBA) transfer amplitude and differential cross-section.

Units throughout: lengths in fm, energies in MeV, and $\hbar c = 197.7\,\mathrm{MeV\,fm}$.

2 Kinematics

2.1 Laboratory–to–CM energy

For a projectile of mass m_1 hitting a target of mass m_2 at rest,

$$E_{\mathrm{cm}} = E_{\mathrm{lab}} \frac{m_2}{m_1 + m_2}. \tag{1}$$

2.2 Reduced mass and mass factor

The two-body reduced mass is

$$\mu = \frac{m_1 m_2}{m_1 + m_2}. \tag{2}$$

Because the radial Schrödinger equation is written in the form $u'' = f(r)\,u$, the convenient combination

$$\xi \equiv \frac{2\mu}{(\hbar c)^2} \tag{3}$$

appears everywhere. In the code this quantity is called `mass-factor` (units: $\mathrm{MeV}^{-1}\,\mathrm{fm}^{-2}$).

2.3 Wave-numbers

The CM wave-number in each channel is

$$k = \sqrt{\xi\, E_{\mathrm{cm}}}, \tag{4}$$

so that $k^2 = \xi E_{\mathrm{cm}}$ with k in fm^{-1} when E_{cm} is in MeV.

2.4 Q-value and exit-channel energy

The Q-value of the reaction $A(a,b)B$ is

$$Q = (m_A + m_a - m_B - m_b)c^2. \tag{5}$$

The exit-channel CM energy is then

$$E_{\rm cm}^{(f)} = E_{\rm cm}^{(i)} + Q. \tag{6}$$

For $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$: masses used are $m_p = 938.272\,\mathrm{MeV}/c^2$, $m_d = 1875.613\,\mathrm{MeV}/c^2$, $m_{16} = 16 \times 931.494\,\mathrm{MeV}/c^2$, $m_{17} = 17 \times 931.494\,\mathrm{MeV}/c^2$, giving $Q \approx 1.92\,\mathrm{MeV}$ and $E_{\rm cm}^{(i)} \approx 17.76\,\mathrm{MeV}$ for $E_{\rm lab}(d) = 20\,\mathrm{MeV}$.

3 Bound-State Wave Function

3.1 Radial Schrödinger equation

The reduced radial wave function $u(r) = r\,\psi(r)$ for the transferred nucleon in the binding potential satisfies

$$-\frac{d^2u}{dr^2} + \left[\frac{l(l+1)}{r^2} + \xi\,(V(r) - E_b)\right] u = 0, \tag{7}$$

where $E_b < 0$ is the binding energy and $V(r)$ is the binding (central + spin-orbit) potential.

3.2 Numerov integration

Equation (7) is solved with the Numerov algorithm. Writing $f_n \equiv f(r_n)$ for the term in square brackets of Eq. (7), the recurrence is

$$\left(1 - \frac{h^2}{12} f_{n+1}\right) u_{n+1} = 2u_n - u_{n-1} + \frac{h^2}{12}\left(10 f_n u_n + f_{n-1} u_{n-1}\right), \tag{8}$$

where h is the radial step and $r_n = n\,h$.

The initial conditions near the origin are

$$u(0) = 0, \qquad u(h) = h^{l+1}. \tag{9}$$

3.3 Binding-potential

The central Woods-Saxon (WS) potential is

$$V_{\rm WS}(r) = \frac{-V_0}{1 + \exp\left(\dfrac{r-R}{a}\right)}, \tag{10}$$

with radius $R = r_0 A^{1/3}$. The Thomas spin-orbit potential adds

$$V_{\rm SO}(r) = V_{so}\,(\boldsymbol{l}\cdot\boldsymbol{s})\,\frac{1}{r}\frac{df_{\rm so}}{dr}, \tag{11}$$

where $f_{\rm so}(r)$ is the WS form factor for the SO geometry and the expectation value of $\boldsymbol{l}\cdot\boldsymbol{s}$ for a nucleon with spin $s = \frac{1}{2}$ and total angular momentum j is

$$\langle \boldsymbol{l}\cdot\boldsymbol{s}\rangle = \frac{j(j+1) - l(l+1) - 3/4}{2}. \tag{12}$$

3.4 Normalization

The bound-state wave function is normalized by

$$\int_0^\infty |u(r)|^2\, dr = 1, \tag{13}$$

evaluated with the composite Simpson rule,

$$\int_0^\infty |u|^2\, dr \approx \frac{h}{3}\left[|u_0|^2 + |u_N|^2 + 4\sum_{n=1,3,5,\ldots} |u_n|^2 + 2\sum_{n=2,4,6,\ldots} |u_n|^2\right]. \tag{14}$$

The reduced radial wave function u and the full radial part $R(r) = u(r)/r$ are related by $\int_0^\infty |R(r)|^2 r^2\, dr = 1$.

4 Optical-Model Potential

The complex optical potential has the general form

$$U(r) = V_{\mathrm{WS}}(r) - i\, W_{\mathrm{vol}}(r) - i\, W_{\mathrm{surf}}(r) + U_{\mathrm{SO}}(r) + V_C(r), \tag{15}$$

with real depths defined as positive.

4.1 Volume terms

Real and imaginary volume WS:

$$V_{\mathrm{WS}}(r) = \frac{-V_0}{1 + \exp\left(\frac{r - R_V}{a_V}\right)}, \tag{16}$$

$$W_{\mathrm{vol}}(r) = \frac{-W_0}{1 + \exp\left(\frac{r - R_W}{a_W}\right)}. \tag{17}$$

4.2 Surface imaginary term

The derivative (surface) imaginary potential is

$$W_{\mathrm{surf}}(r) = -4\, W_D\, f_S(r)[1 - f_S(r)]\,, \quad f_S(r) = \frac{1}{1 + \exp\left(\dfrac{r - R_S}{a_S}\right)}, \tag{18}$$

which is proportional to $4a_S\, df_S/dr$ and peaks at $r \approx R_S$.

4.3 Spin-orbit term

$$U_{\mathrm{SO}}(r) = V_{so}\,(\boldsymbol{l}\cdot\boldsymbol{s})\,\frac{1}{r}\frac{df_{so}(r)}{dr}, \qquad f_{so}(r) = \frac{1}{1 + \exp\left(\dfrac{r - R_{so}}{a_{so}}\right)}. \tag{19}$$

4.4 Coulomb potential

For projectile charge $Z_1 e$ incident on a uniformly charged sphere of charge $Z_2 e$ and radius R_C:

$$V_C(r) = \begin{cases} \dfrac{Z_1 Z_2 e^2}{r} & r > R_C, \\ \dfrac{Z_1 Z_2 e^2}{R_C^2}\, r & r \le R_C. \end{cases} \tag{20}$$

The Coulomb constant is $e^2 = 1.44\,\mathrm{MeV\,fm}$.

4.5 Parameters for $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$

Table 1 lists the handbook parameters used for both channels ($A^{1/3}$ scalings are $16^{1/3}$ and $17^{1/3}$, respectively).

Table 1: Optical-model parameters (Table 5.1 / 5.2 from the handbook) at 10 MeV/u.

Parameter	$d + {}^{16}\mathrm{O}$ (entrance)	$p + {}^{17}\mathrm{O}$ (exit)
V_0 (MeV)	88.955	49.544
$r_{0,V}$ (fm)	1.149	1.146
a_V (fm)	0.751	0.675
W_0^{vol} (MeV)	2.348	2.061
$r_{0,W}$ (fm)	1.345	1.146
a_W (fm)	0.603	0.675
W_D (MeV)	10.218	7.670
$r_{0,S}$ (fm)	1.397	1.302
a_S (fm)	0.687	0.528
V_{so} (MeV)	3.557	5.296
$r_{0,so}$ (fm)	0.972	0.934
a_{so} (fm)	1.011	0.590
$r_{0,C}$ (fm)	1.303	1.419

5 Distorted Waves

5.1 Radial equation and Numerov integration

The reduced distorted wave $u(r) = r\,\chi(r)$ satisfies the complex Schrödinger equation

$$-\frac{d^2u}{dr^2} + \left[\frac{l(l+1)}{r^2} + \xi(U(r) - E_{\mathrm{cm}})\right] u = 0. \tag{21}$$

The Numerov recurrence (8) is applied with complex f_n derived from the complex optical potential $U(r)$.

5.2 Normalization modes

The code implements four post-integration normalization conventions.

Raw (`:raw`) No rescaling; relative amplitudes between partial waves are preserved.

Coulomb-tail (`:coulomb-tail`) Scale u so that $|u(r_{\max})| = |H_L^+(kr_{\max}, \eta)|$ where H_L^+ is the outgoing Coulomb–Hankel function:

$$u_i \longleftarrow u_i \times \frac{|H_L^+(kr_{\max}, \eta)|}{|u(r_{\max})|}. \tag{22}$$

BIND-flux (`:bind-flux`) — DWUCK4-style This mode reproduces the DWUCK4 `BIND` normalization. Define

$$\text{SCALE} = |H_L^-(kr_N, \eta)|, \tag{23}$$

where $r_N = N\,h$ is the last grid point. Then each grid value is rescaled as

$$u(r_i) \longleftarrow u(r_i)\,\frac{\text{SCALE}}{k\,r_i}, \qquad i \geq 1, \quad u(0) = 0. \tag{24}$$

The effect is that in the outer region $u(r) \approx H_L^-(kr, \eta)/(kr)$.

6 S-Matrix and Phase Shifts

6.1 R-matrix

From the Numerov solution at a matching radius a, the logarithmic derivative (R-matrix element times a) is

$$R \equiv \frac{u(a)}{a\,u'(a)}, \tag{25}$$

computed either from the real-valued stepping (Coulomb + real WS) or from the complex Numerov stepping (Coulomb + complex WS) via `r-matrix-complex-imag-ws`.

6.2 Nuclear S-matrix

Matching u to incoming (H_L^-) and outgoing (H_L^+) Coulomb Hankel functions at radius a:

$$S_L^n = \frac{H_L^-(\rho) - \rho R\,\partial_\rho H_L^-(\rho)}{H_L^+(\rho) - \rho R\,\partial_\rho H_L^+(\rho)}, \qquad \rho = k\,a. \tag{26}$$

The Coulomb Hankel functions at arbitrary η are evaluated using the asymptotic expansion in terms of the Tricomi confluent hypergeometric function $U(a, b; z)$:

$$H_L^\pm(\eta, \rho) = e^{\pm i\theta_L}\,(-2i\rho)^{L+1\mp i\eta}\,U(L+1\pm i\eta,\, 2(L+1);\, \mp 2i\rho)\,, \tag{27}$$

where $\theta_L = \rho - \frac{L\pi}{2} + \sigma_L - \eta\ln(2\rho)$.

6.3 Coulomb phase shifts

The Coulomb phase shift is

$$\sigma_L(\eta) = \arg\,\Gamma(L+1+i\eta). \tag{28}$$

The relative phase factor $e^{2i(\sigma_L-\sigma_0)}$ is computed without Gamma-function evaluations using the product formula

$$e^{2i(\sigma_L-\sigma_0)} = \prod_{k=1}^{L} \frac{k+i\eta}{k-i\eta}, \tag{29}$$

each factor being a pure phase of magnitude 1 (implemented in `functions/coulomb-phase-diff`).

6.4 Sommerfeld parameter

The Sommerfeld parameter for CM scattering is

$$\eta = \frac{Z_1 Z_2 e^2 \cdot \xi}{2k} = \frac{Z_1 Z_2 e^2}{2(\hbar c)^2}\,\frac{\hbar^2 \mu}{k}. \tag{30}$$

7 Elastic Scattering

7.1 Nuclear partial-wave bracket

From Thompson & Nunes, Eq. (3.1.88), the nuclear contribution to the scattering amplitude is built from

$$e^{2i\sigma_L}(S_L^n - 1)\,, \tag{31}$$

which vanishes for $L > L_{\max}$ (beyond the nuclear range).

7.2 Total elastic amplitude

$$f(\theta) = f_C(\theta) + f_N(\theta), \tag{32}$$

where the Coulomb amplitude is

$$f_C(\theta) = -\frac{\eta}{2k\sin^2(\theta/2)}\,\exp\!\big[i\big(-\eta\ln\sin^2(\theta/2) + 2\sigma_0\big) + i\pi\big]\,, \tag{33}$$

and the nuclear partial-wave amplitude is

$$f_N(\theta) = -\frac{i}{k}\sum_{L=0}^{L_{\max}} (2L+1)\,P_L(\cos\theta)\,e^{2i\sigma_L}(S_L^n - 1)\,. \tag{34}$$

Removing the overall phase $e^{2i\sigma_0}$ from both amplitudes defines $\tilde{f}_C$ and $\tilde{f}_N$, which satisfy $|f_C + f_N|^2 = |\tilde{f}_C + \tilde{f}_N|^2$:

$$\tilde{f}_C(\theta) = -\frac{\eta}{2k\sin^2(\theta/2)}\,\exp\!\big[-i\eta\ln\sin^2(\theta/2) + i\pi\big]\,, \tag{35}$$

$$\tilde{f}_N(\theta) = -\frac{i}{k}\sum_{L=0}^{L_{\max}} (2L+1)\,P_L(\cos\theta)\,e^{2i(\sigma_L-\sigma_0)}(S_L^n - 1)\,. \tag{36}$$

7.3 Differential cross-section

$$\frac{d\sigma}{d\Omega} = |f_C + f_N|^2 \quad \big[\,\mathrm{fm}^2/\,\mathrm{sr}\big]\,. \tag{37}$$

The code converts to mb/sr via the factor $1\,\mathrm{fm}^2 = 10\,\mathrm{mb}$.

8 DWBA Transfer Amplitude: Zero-Range Approximation

8.1 Zero-range D0 approximation

In the zero-range (ZR) DWBA the inter-cluster binding interaction $V_{an}(r_{an})$ is replaced by a contact term:

$$\langle \mathbf{r}_\beta | \, V_{an} \, | \phi_d \rangle \;\approx\; D_0 \, \delta^{(3)}(\mathbf{r}_\alpha - \mathbf{r}_\beta), \tag{38}$$

with the zero-range constant $D_0 = -122.4\,\mathrm{MeV\,fm}^{3/2}$ for (d,p) reactions.

8.2 Form factor

The bound-state form factor entering the ZR integral is

$$F_{\ell sj}(r) = R_n(r) \equiv \frac{u_n(r)}{r}, \tag{39}$$

where $u_n(r)$ is the normalized reduced bound-state wave function of the transferred neutron in ^{17}O.

8.3 ZR radial integral

Following Austern (1970), Eq. (5.5), the ZR radial matrix element is

$$I^{\ell sj}_{L_\beta L_\alpha} = \frac{M_B}{M_A} \frac{4\pi}{k_\alpha k_\beta} \int_0^\infty F_{\ell sj}(r) \, \chi^{(+)}_\alpha(L_\alpha, r) \, \chi^{(-)*}_\beta\!\left(L_\beta, \, \frac{M_A}{M_B} r\right) r^2 \, dr, \tag{40}$$

where $M_A = M(^{16}\mathrm{O})$, $M_B = M(^{17}\mathrm{O})$, and $\chi^{(\pm)}_{\alpha(\beta)}$ are the distorted waves (reduced: u/r). The ZR mass-ratio M_A/M_B gives the argument scaling of the exit distorted wave.

The integral is evaluated with the composite Simpson rule.

8.4 Angular-momentum selection rules

For a fixed transferred orbital angular momentum ℓ, the admissible (L_α, L_β) pairs satisfy:

- parity: $(-1)^{L_\alpha} = (-1)^{L_\beta + \ell}$,
- triangle inequality: $|L_\alpha - \ell| \le L_\beta \le L_\alpha + \ell$,
- $L_\alpha, L_\beta \le L_{\max}$.

8.5 Reduced amplitude (Austern Eq. 5.6)

The reduced transfer amplitude for magnetic quantum number m is

$$\beta_{\ell m}(\Theta) = \sum_{L_\alpha, L_\beta} i^{L_\alpha - L_\beta - \ell} \, e^{i(\sigma_{\alpha L_\alpha} + \sigma_{\beta L_\beta})} \sqrt{2L_\beta + 1} \, I^{\ell sj}_{L_\beta L_\alpha} \, \langle L_\beta \, \ell; \, 0\,0 \mid L_\alpha \, 0 \rangle \, \langle L_\beta \, \ell; \, -m \, m \mid L_\alpha \, 0 \rangle \, Y^{-m}_{L_\beta}(\Theta, 0), \tag{41}$$

where $\sigma_{\alpha L_\alpha} = \sigma_{L_\alpha}(\eta_\alpha)$ and $\sigma_{\beta L_\beta} = \sigma_{L_\beta}(\eta_\beta)$ are the Coulomb phase shifts in the entrance and exit channels, Θ is the CM scattering angle, and $Y^{-m}_{L_\beta}(\Theta, 0)$ is a spherical harmonic evaluated at azimuthal angle 0 (coplanar kinematics).

8.6 Transfer amplitude

In the ZR approximation the full transfer amplitude is

$$T_m(\Theta) = D_0 \sqrt{2\ell+1}\, \beta_{\ell m}(\Theta). \tag{42}$$

8.7 Differential cross-section

The differential cross-section for an unpolarized beam (spin-average over initial states, spin-sum over final states) is

$$\frac{d\sigma}{d\Omega}(\Theta) = \frac{\mu_i\,\mu_f}{(2\pi\hbar^2)^2}\,\frac{k_f}{k_i}\,\mathcal{S}\,\sum_m |T_m(\Theta)|^2, \tag{43}$$

or equivalently, using the mass factors $\xi_i = 2\mu_i/(\hbar c)^2$:

$$\frac{d\sigma}{d\Omega}(\Theta) = \frac{\xi_i\,\xi_f}{16\pi^2}\,\frac{k_f}{k_i}\,\mathcal{S}\,\sum_m |T_m|^2 \quad \times 10 \quad [\,\mathrm{mb/\,sr}]. \tag{44}$$

The factor of 10 converts $\mathrm{fm}^2 \to \mathrm{mb}$ ($1\,\mathrm{fm}^2 = 10\,\mathrm{mb}$).

The spectroscopic factor $\mathcal{S}$ absorbs spin-statistics:

$$\mathcal{S} = S_{\text{factor}} \times \underbrace{\frac{2J_f+1}{2J_i+1}}_{\text{nuclear spin}} \times \underbrace{\frac{1}{3}}_{\text{deuteron spin}}. \tag{45}$$

For $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$: $J_i = 0$, $J_f = 5/2$, giving $(2J_f+1)/(2J_i+1) = 6$.

The sum over m runs over $m = -\ell, \ldots, \ell$ with each term $|T_m|^2 = D_0^2\,(2\ell+1)\,|\beta_{\ell m}|^2$.

9 Specific Reaction: $^{16}\mathrm{O}(d,p)^{17}\mathrm{O}$

9.1 Channels

- **Entrance** (α): $d + {}^{16}\mathrm{O}$, $E_{\text{lab}}(d) = 20\,\mathrm{MeV}$, $k_\alpha = \sqrt{\xi_i E_{\text{cm}}^{(i)}}$.
- **Exit** (β): $p + {}^{17}\mathrm{O}(\text{g.s.})$, $k_\beta = \sqrt{\xi_f E_{\text{cm}}^{(f)}}$.

9.2 Transferred nucleon

A neutron is transferred to the $1d_{5/2}$ orbit of $^{17}\mathrm{O}$: $n = 1$, $\ell = 2$, $s = 1/2$, $j = 5/2$. The binding energy is $E_b \approx -4.14\,\mathrm{MeV}$ (bound-state well is fit to reproduce this value using a WS potential with $R_n = 2.85\,\mathrm{fm}$, $a_n = 0.60\,\mathrm{fm}$).

9.3 Partial-wave summation

$L_{\max}$ is chosen such that the nuclear amplitude has converged. For this reaction, $L_{\max} \approx$ 15–20.

9.4 Summary of the calculation chain

1. Solve the kinematics: $E_{\rm cm}^{(i)}$, $E_{\rm cm}^{(f)}$, k_α, k_β, η_α, η_β.
2. Solve the bound-state equation for $u_n(r)$; normalize it.
3. For each $L_\alpha \in [0, L_{\rm max}]$:
   (a) Solve the distorted-wave equation for $\chi_\alpha^{(+)}$ with the d+^{16}O optical potential.
4. For each L_β (selected by the triangle rule):
   (a) Solve the distorted-wave equation for $\chi_\beta^{(-)}$ with the p+^{17}O optical potential.
5. Compute $I_{L_\beta L_\alpha}$ via Eq. (40) (Simpson quadrature).
6. Attach Coulomb phases: $I_{L_\beta L_\alpha} \xrightarrow{\sigma_\alpha, \sigma_\beta}$ `handbook-zr-rows-with-coulomb-sigma`.
7. Compute $\beta_{\ell m}(\Theta)$ via Eq. (41) for $m = -\ell, \ldots, \ell$.
8. Accumulate $\sum_m |T_m|^2$ and apply Eq. (44).

10 Numerical Parameters

Table 2: Default numerical parameters.

Parameter	Value	Description
h	0.05 fm	Radial step size
$r_{\rm max}$	100 fm	Radial integration limit
$L_{\rm max}$	20	Maximum partial wave
D_0	$-122.4\,\mathrm{MeV\,fm}^{3/2}$	Zero-range constant for (d,p)
$\hbar c$	197.7 MeV fm	Conversion constant
e^2	1.44 MeV fm	Coulomb constant

11 Code–Formula Correspondence

References

1. N. Austern, *Direct Nuclear Reaction Theories*, Wiley–Interscience, 1970. [Eqs. (4.60), (5.3)–(5.6).]
2. I. J. Thompson and F. M. Nunes, *Nuclear Reactions for Astrophysics*, Cambridge University Press, 2009. [Eqs. (3.1.81), (3.1.84), (3.1.88).]
3. *Handbook of direct nuclear reaction for retarded theorist*, internal reference. [Tables 5.1, 5.2, § 5.5.2.]
4. P. D. Kunz, DWUCK4 code manual (unpublished). [BIND normalization convention.]

Table 3: Mapping between key formulas and code functions.

Formula	Equation	Code function
Lab-to-CM energy	Eq. (1)	`functions/lab-to-cm-energy`
Mass factor ξ	Eq. (3)	`functions/mass-factor-from-mu`
Wave-number k	Eq. (4)	`(Math/sqrt (* mass-factor E))`
Bound-state Numerov	Eq. (8)	`transfer/solve-bound-state-numerov`
Spin-orbit $\boldsymbol{l} \cdot \boldsymbol{s}$	§ 3.3	`transfer/l-dot-s-nucleon`
Normalization	§ 3.4	`transfer/normalize-bound-state`
Optical potential	Eq. (15)	`transfer/optical-potential-woods-saxon`
Surface imaginary	Eq. (18)	(W-surf branch in `optical-potential-woods-saxon`)
Distorted-wave Numerov	Eq. (8)	`transfer/distorted-wave-optical`
BIND normalization	Eq. (24)	`:normalize-mode :bind-flux`
Nuclear S-matrix	Eq. (26)	`functions/s-matrix`
Coulomb phase σ_L	Eq. (28)	`functions/coulomb-sigma-L`
Phase-diff product	Eq. (29)	`functions/coulomb-phase-diff`
Sommerfeld η	Eq. (30)	`functions/channel-sommerfeld-eta`
Coulomb amplitude f_C	Eq. (33)	`functions/coulomb-scattering-amplitude-`...
Nuclear amplitude f_N	Eq. (34)	`functions/elastic-nuclear-amplitude-fn`
Nuclear bracket	Eq. (31)	`functions/partial-wave-exp2sigma-Sn-minus-one`
Form factor $F = u/r$	Eq. (39)	`transfer/handbook-F-lsj-radial-from-neutron-bound-u`
ZR radial integral I	Eq. (40)	`transfer/handbook-radial-integral-I-zr`
Austern amplitude β	Eq. (41)	`transfer/handbook-zr-multipole-amplitude-sum`
Transfer amplitude T_m	Eq. (42)	`(mul pref (handbook-zr-multipole-amplitude-sum ...))`
Differential cross-section	Eq. (44)	`transfer/transfer-differential-cross-section`
Spin prefactor $\mathcal{S}$	Eq. (45)	`transfer/transfer-nuclear-spin-statistical-factor`
		`transfer/transfer-unpolarized-deuteron-spin-factor`